AI Resume Analyzer - Complete Development Roadmap

Project Goal

Build an AI-powered Resume Analyzer that allows users to:

- Upload their resume (PDF)
- Receive an ATS score
- Identify missing skills
- Get resume improvement suggestions
- Match their resume against job descriptions
- Track previous resume analyses

This project will help learn:

- Frontend Development
- Backend APIs
- Authentication
- File Uploads
- AWS Services
- OpenAI APIs
- Database Design
- Cloud Deployment
- Product Development

Final Product Features

User Features

- Sign Up / Login
- Upload Resume
- Analyze Resume
- View ATS Score
- View Skill Gap Analysis
- Resume History
- Job Description Matching
- Download Analysis Report

Technology Stack

Frontend

React + Vite

Reason:

- Fast development
- Industry standard
- Excellent ecosystem

Tailwind CSS

Reason:

- Rapid UI development
- Modern responsive design

Backend

Node.js

Express.js

Reason:

- Matches existing JavaScript skills
- Large ecosystem
- Easy API development

Database

PostgreSQL

Hosted on:

Supabase

Reason:

- Free tier available
- Production ready

- Easy authentication integration

AI Layer

OpenAI API

Initial model:

- GPT-4o

Future:

- Claude
- Gemini

Authentication

Supabase Auth

Features:

- Email Login
- Password Login
- Social Login (Optional)

File Storage

AWS S3

Purpose:

- Store uploaded resumes
- Secure file access
- Scalable storage

Deployment

Frontend:

- Vercel

Backend:

- Render

Database:

- Supabase

Storage:

- AWS S3

System Architecture

User | React Frontend | Express Backend API | Resume Upload | AWS S3 | PDF Parser | OpenAI API | PostgreSQL Database | Analysis Dashboard

---

Development Roadmap

Phase 1: Environment Setup

Install Required Tools

1. Node.js
2. VS Code
3. Git
4. GitHub Account

VS Code Extensions

- ESLint
- Prettier
- GitHub Copilot
- Thunder Client

Phase 2: Frontend Setup

Create Project

Command:

```
npm create vite@latest ai-resume-analyzer
```

Select:

- React
- JavaScript

Install Dependencies:

npm install

Run Project:

npm run dev

Expected Output:

http://localhost:5173

Phase 3: Backend Setup

Create Folder:

backend/

Initialize:

npm init -y

Install Dependencies:

npm install express cors dotenv multer

Create:

server.js

Run Server:

node server.js

Expected Output:

http://localhost:5000

Phase 4: Resume Upload Feature

Install

npm install multer

Create API

POST /upload

Functionality

- Upload PDF
- Validate file type
- Store temporarily
- Return upload status

Phase 5: AWS S3 Integration

AWS Setup

1. Create AWS Account
2. Create S3 Bucket

Example:

resume-analyzer-storage

Install SDK

npm install @aws-sdk/client-s3

Implement

- Upload PDF
- Generate file URL
- Store URL in database

Phase 6: Resume Parsing

Install

npm install pdf-parse

Process

PDF → Extract Text → Clean Text → Send to AI

Example Output:

Name: Sheshi Kiran

Skills: JavaScript Angular AWS GraphQL

Experience: Software Developer

Phase 7: OpenAI Integration

Install

npm install openai

Create OpenAI Account

Generate API Key

Store in:

.env

OPENAI_API_KEY=your_key

Phase 8: Resume Analysis Engine

AI Prompt

You are an ATS expert.

Analyze this resume and return:

1. ATS Score
2. Missing Skills
3. Strengths
4. Weaknesses
5. Improvement Suggestions

Resume:

Expected Response

ATS Score: 82

Strengths:

- AWS
- Angular
- GraphQL

Missing Skills:

- Docker
- Kubernetes

Suggestions:

- Add measurable achievements
- Improve project descriptions

Phase 9: Database Design

Users Table

Fields:

- id
- email
- password
- created_at

Resumes Table

Fields:

- id
- user_id
- resume_url
- analysis
- created_at

Phase 10: Dashboard Development

Dashboard Features

- Profile Section
- Resume History
- ATS Score History
- Analysis Results
- Suggested Improvements

Phase 11: Authentication

Features

- User Registration
- Login
- Logout
- Protected Routes

Using:

Supabase Auth

Phase 12: Job Description Matching

Feature

User Pastes:

- Job Description

System Compares:

Resume VS Job Description

Returns:

- Match Percentage
- Missing Skills
- Keyword Gap
- Recommendations

Example:

Match Score: 78%

Missing:

- Docker
- CI/CD
- Kubernetes

Phase 13: Deploy to Production

Frontend Deployment

Platform: Vercel

Steps:

1. Push Code to GitHub
2. Connect GitHub to Vercel
3. Deploy

Backend Deployment

Platform: Render

Steps:

1. Push Backend to GitHub
2. Connect Repository
3. Add Environment Variables
4. Deploy

Database Deployment

Platform: Supabase

Steps:

1. Create Project
2. Create Tables
3. Connect Backend

Storage Deployment

Platform: AWS S3

Steps:

1. Create Bucket
2. Configure IAM
3. Connect Backend

Recommended Folder Structure

ai-resume-analyzer/

frontend/ ├── src/ ├── pages/ ├── components/ ├── services/ ├── hooks/

backend/ ├── routes/ ├── controllers/ ├── services/ ├── middleware/ ├── utils/ ├── uploads/ ├── server.js

database/ ├── schema.sql

docs/ ├── architecture.md

6-Week Execution Plan

Week 1

- React Setup
- Tailwind Setup
- Backend Setup
- GitHub Repository

Week 2

- Resume Upload
- AWS S3 Integration

Week 3

- PDF Parsing
- OpenAI Integration

Week 4

- ATS Analysis
- Database Setup

Week 5

- Authentication
- Dashboard Development

Week 6

- Job Description Matching
- Deployment
- Portfolio Documentation
- LinkedIn Project Showcase

Project Outcome

By completing this project, you will gain hands-on experience in:

- React
- Node.js
- REST APIs
- AWS S3
- PostgreSQL
- Supabase
- OpenAI APIs
- Authentication
- Cloud Deployment
- Product Development

This project can become a strong portfolio piece and demonstrate full-stack AI product development skills to recruiters, clients, and startup founders.